CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
FACULTY OF MANAGEMENT STUDIES (FMS)
BBA PROGRAMME
SEMESTER – I, UNIVERSITY EXAMINATION, MARCH - 2018
BM104 BUSINESS MATHEMATICS (OLD)
BM104.1 BUSINESS MATHEMATICS (NEW)

Date: 29/03/2018, Thursday **Time**: 10.00 AM to 10.30 AM **Total Marks**: 15

Student's ID No.									

Student's Name: ____________________

Student's Sign : ____________________

Supervisor's Name : ____________________

Supervisor's Sign: ____________________

Important Instructions:

- All questions are compulsory.
- This is a closed book examination.
- There are two sections in the question paper. Section – I contains Q. No.: 1 which is of 15 marks and Section – II contains total five questions wherein Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.
- Section – I should be answered in the given question paper and Section – II should be answered in a separate answer book.
- Maximum time allotted for Section – I is 30 minutes. Answer Book for Section – II will be given when the Question Paper for Section – I is returned. However, candidates who complete Section – I earlier than 30 minutes, can collect Answer Book for Section – II immediately after handing over Section – I question paper.
- Figures on the right indicate marks for each question.

SECTION – I

Q.1 Objective Type Questions: **15**

A) Match the following: (i – vi) **06**

Q. No.:		Probable Answer Keys		Write Your Answer Here
i.	(A')'	a)	Matrix	**i. -**
ii.	Order of the power set	b)	Algebra	**ii. -**
iii.	Words can be formed by using all the letters of word - 'ALIVE'	c)	0	**iii. -**
iv.	Array of elements in rectangular arrangement	d)	$n * a * x^{n-1}$	**iv. -**
v.	$\frac{dy}{dx}$ $Y = ax^n$	e)	5^2	**v. -**
vi.	Lim (x,y)-> (0,0) $\left\{\frac{x3-y3}{x-y}\right\}$	f)	5!	**vi. -**
		g)	U	
		h)	$n * x^{n-1}$	
		i)	A	
		j)	2^n	

B) Multiple-Choice Questions (vii – ix) ***(Note: Encircle the correct answer)*** **03**

vii. ***In how many ways a committee, consisting of 5 men and 6 women can be formed from 8 men and 10 women?***

a) 266 b) 5040

c) 11760 d) 86400

viii. ***Power set of empty set has exactly __________ subset.***

a) One b) Two

c) Three d) Four

ix. **In square matrix, all elements along primary diagonal are**

a) One b) Zero

c) Three d) Two

C) Fill-in the Blanks (x – xii) (*You may choose from the following answers):* **03**

$30C_2$	Infinite	30!	Transpose	Addition	Finite

x. The set of positive integers is ______________

xi. There are 30 people in a group. If all shake hands with one another, _______________________handshakes are possible.

xii. In _______________________________ of matrix A, columns of matrix A becomes rows.

D) True or False (xiii – xv) ***(Note: Encircle the correct answer)*** **03**

xiii. A box contains 4 black, 3 red and 6 green marbles. 2 marbles are drawn from the box at random. The probability that both the marbles are of the same color is 24/78.

(True) or **(False)**

xiv. The Cartesian Product B x A is equal to the Cartesian product A x B.

(True) or **(False)**

xv. Unit matrix written in format of square matrix is also called as identity matrix

(True) or **(False)**

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Date: 29/03/2018, Thursday **Time**: 10.30 AM to 01.00 PM **Total Marks: 55**

Note: - (i) All questions are compulsory.
(ii) This is a closed book examination.
(iii) There are two sections in the question paper. **Section – I** contains Q. No.: 1 of 15 marks and **Section – II** contains total five questions where in Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.
(iv) **Section – I** should be answered in the given question paper and **Section – II** should be answered in a separate answer book.
(v) Maximum time allotted for **Section – II** is 150 minutes.
(vi) Figures on the right indicate marks for each question.

SECTION – II

Q.2 **In how many ways can the letters of the following words be arranged: *(words may not have any meaning)*** **10**

i. 'MATHEMATICS'
ii. 'DETAIL' *(arrange in such a way that the vowels must occupy only the odd positions)*
iii. ' JUDGE' *((arrange in such a way that the vowels come together)*
iv. 'BIHAR'
v. 'DELHI' *(using each letter exactly once)*

OR

Q.2 ***Solve the following problems (A and B):*** **10**

A There are 6 periods in each working day of a school. In how many ways can one organize 5 subjects such that each subject is allowed at least one period? [05]

B 25 buses are running between two places P and Q. In how many ways can a person go from P to Q and return by a different bus? [05]

Q.3 ***Find the limit of the following functions (A and B):*** **10**

A $\lim_{x \to 2} \dfrac{x^2+4x-12}{x^2-2x}$ [05]

B $\lim_{x \to (-4)} \dfrac{\frac{1}{4}+\frac{1}{x}}{4+x}$ [05]

Q.4 ***Solve the following problems (A and B):*** **10**

A Let A = {a, b, d, e}, B = {b, c, e, f} and C = {d, e, f, g} , verify [05]
i. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
ii. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

B Given three sets A, B and C such that: A = {x : x is a natural number between 10 [05] and 16}, B = {set of even numbers between 8 and 20} and C = {7, 9, 11, 14, 18, 20}.

Find the difference of sets using Venn diagram:

(i) A – B (ii) B – C (iii) C – A (iv) B – A (v) A - C

OR

Q.4 ***Solve the following problems (A and B):*** **10**

A Evaluate $\int \left(x - \frac{1}{x}\right)^2 \, dx$ [05]

B Evaluate $\int (4x^3 - 1) \, dx$ [05]

Q.5 Differentiate the following with respect to x : **10**

A $6X^4 - 7X^3 + 3X^2 - X + 8$ [05]

B $\frac{2}{3} X^{2/3}$ [05]

Q.6 **Case-let** **15**

Two shops A and B have in stock the following brand of tube lights

Shops	**Brand**		
	Bajaj	**Philips**	**Surya**
Shop A	43	62	36
Shop B	24	18	60

Shop A places order for 30 Bajaj, 30 Philips, and 20 Surya brand of tube lights, whereas shop B orders 10, 6, 40 numbers of the three varieties. Due to the various factors, they receive only half of the order as supplied by the manufacturers. The cost of each tube lights of the three types are Rs. 42, Rs. 38 and Rs. 36 respectively.

Represent the following as matrices

(i) Initial stock
(ii) the order
(iii) the supply
(iv) final stock
(v) cost of individual items (column matrix)
(vi) total cost of stock in the shops.